Independent Validation of Energy-Flow Cosmology BAO Predictions: Cross-Survey Transfer from DESI DR2 to BOSS DR12

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Abstract

We present an independent validation of Energy-Flow Cosmology (EFC) using Baryon Acoustic Oscillation (BAO) measurements from BOSS DR12, testing parameter transfer from DESI DR2. Using canonical regime gating where $z < z_{L1L2}$ activates the L2 modification, we find that DESI best-fit parameters ($z_{L1L2} = 1.01$, $\alpha_{L2} = 0.045$) reduce the BOSS χ^2 from 20.83 to 13.06 ($\Delta\chi^2 = -7.77$) without refitting. This improvement is robust: it persists under diagonal covariance ($\Delta\chi^2 = -4.24$), and eigenmode analysis confirms that gains arise primarily from strongly-penalized modes rather than covariance exploitation. Combined BOSS+eBOSS data yield best-fit $z_{L1L2} \approx 1.6$ and $\alpha_{L2} \approx 0.036$, consistent with DESI constraints. These results provide independent support for EFC modifications to the Hubble rate at the 4–5% level for $z < z_{L1L2}$.

1 Introduction

Energy-Flow Cosmology (EFC) proposes modifications to the standard ΛCDM expansion history through a regime-dependent correction to the Hubble rate [1]. Recent analysis of DESI DR2 BAO data suggested preference for EFC with transition redshift $z_{L1L2} \approx 1.01$ and L2 amplitude $\alpha_{L2} \approx 0.045$ [1].

A critical test of any cosmological model is whether parameters derived from one dataset successfully predict observations in independent surveys. In this work, we perform such a transfer test using BOSS DR12 consensus BAO measurements [2], which are completely independent of DESI in instrumentation, analysis pipeline, and systematic error budget.

2 EFC Model and Regime Gating

The EFC modification to the Hubble rate is implemented through a smooth transition function:

$$H_{\mathrm{EFC}}(z) = H_{\Lambda\mathrm{CDM}}(z)\,[1 + \alpha_{L2} \cdot \Theta(z)] \tag{1}$$

where the gating function is

$$\Theta(z) = \frac{1}{2}\left[1 + \tanh\left(\frac{z_{L1L2} - z}{\Delta z}\right)\right] \tag{2}$$

with transition width $\Delta z = 0.05$.

The canonical interpretation is:

- $z < z_{L1L2}$: $\Theta \approx 1 \to$ L2 regime active
- $z > z_{L1L2}$: $\Theta \approx 0 \to$ L1 (ΛCDM) regime

For DESI best-fit $z_{L1L2} = 1.01$, all BOSS data points ($z = 0.38$–0.61) lie in the L2 regime, meaning DESI parameters predict a $\sim$4.5% modification to the Hubble rate throughout the BOSS redshift range.

3 Data and Methodology

3.1 BOSS DR12 Consensus BAO

We use the BOSS DR12 consensus BAO measurements from Ref. [2], comprising D_M/r_s and D_H/r_s at three effective redshifts ($z_{\rm eff} = 0.38, 0.51, 0.61$), for a total of $N = 6$ data points. The full 6×6 correlation matrix is employed throughout.

For the ΛCDM baseline $\chi^2_{\Lambda\rm CDM}$ we adopt the Planck 2018 best-fit parameters [4] ($H_0 = 67.4$ km/s/Mpc, $\Omega_m = 0.315$, $r_s = 147.09$ Mpc), with EFC disabled ($\alpha_{L2} = 0$). This ensures the transfer test isolates the effect of the late-time EFC modification rather than changing the baseline calibration.

We model the BAO observables as $D_H(z)/r_s$ and $D_M(z)/r_s$, where $D_H(z) \equiv c/H(z)$ is the Hubble distance and $D_M(z)$ is the comoving angular diameter distance. Since EFC modifies only the late-time expansion history, we treat the sound horizon r_s as unchanged relative to the baseline early-universe physics. The EFC modification enters as a *multiplicative* change to $H(z)$: BAO responds via $D_H \propto 1/H$ (direct) and $D_M \propto \int dz/H$ (integral), explaining why a modest $\sim$4% change in H produces measurable $\Delta\chi^2$.

3.2 Transfer Test Protocol

The transfer test evaluates DESI-derived parameters on BOSS data *without refitting*:

1. Compute $\chi^2_{\Lambda\rm CDM}$ on BOSS using standard cosmology
2. Compute $\chi^2_{\rm EFC}$ on BOSS using DESI parameters ($z_{L1L2} = 1.01$, $\alpha_{L2} = 0.045$)
3. Report $\Delta\chi^2 = \chi^2_{\rm EFC} - \chi^2_{\Lambda\rm CDM}$

Since no parameters are fitted, $k_{\rm eff} = 0$ for this comparison, and $\Delta\chi^2$ is the relevant statistic.

3.3 Covariance Diagnostics

To ensure the χ^2 improvement is genuine and not a covariance artifact, we perform three diagnostic tests:

Test 1 (Whitening): Transform residuals via Cholesky decomposition $\mathbf{C} = \mathbf{L}\mathbf{L}^T$, defining whitened residuals $\mathbf{w} = \mathbf{L}^{-1}\mathbf{r}$ such that $\chi^2 = \mathbf{w}^T\mathbf{w} = \sum_i w_i^2$.

Test 2 (Eigenmode): Decompose $\mathbf{C} = \mathbf{U}\,\mathrm{diag}(\lambda)\,\mathbf{U}^T$ and compute mode contributions $\chi^2 = \sum_i a_i^2/\lambda_i$ where $\mathbf{a} = \mathbf{U}^T\mathbf{r}$.

Test 3 (Robustness): Interpolate between diagonal and full covariance: $\mathbf{C}(\rho) = (1-\rho)\,\mathrm{diag}(\mathbf{C}) + \rho\,\mathbf{C}$ for $\rho \in [0, 1]$.

4 Results

4.1 Transfer Test

Table 1 presents the transfer test results. DESI parameters reduce the BOSS χ^2 by 7.77, a substantial improvement achieved without any parameter adjustment.

Table 1: Transfer test: DESI parameters on BOSS data (no refit).

Model	χ^2	Notes
ΛCDM	20.83	Baseline
EFC (DESI params)	13.06	$z_{L1L2} = 1.01$, $\alpha_{L2} = 0.045$
$\Delta\chi^2$	-7.77	Transfer supported

4.2 Covariance Diagnostics

Whitening analysis (Table 2) reveals that component w_5 dominates the improvement with $\Delta\chi^2 = -11.39$. EFC improves on 3 of 6 whitened components, but the improvement in strongly-constrained components outweighs the degradation in weakly-constrained ones.

Table 2: Whitened residual component χ^2 contributions. Components w_i are obtained via Cholesky decomposition of the covariance matrix; these are not eigenmodes but decorrelated linear combinations of the original residuals.

Component	$\chi^2_{\Lambda\mathrm{CDM}}$	χ^2_{EFC}	$\Delta\chi^2$
w_1	1.27	2.29	+1.01
w_2	1.09	2.14	+1.05
w_3	0.01	2.82	+2.81
w_4	1.50	0.48	-1.02
w_5	15.30	3.91	$\mathbf{-11.39}$
w_6	1.65	1.43	-0.23
Total	20.83	13.06	-7.77

Eigenmode analysis shows that EFC gains $\Delta\chi^2 = -5.50$ from low-λ (strongly-penalized) modes versus $\Delta\chi^2 = -2.27$ from high-λ (weakly-penalized) modes. This confirms the improvement is genuine rather than exploiting weakly-constrained directions.

Robustness sweep (Fig. 1) demonstrates that EFC outperforms ΛCDM across all covariance interpolations:

- $\rho = 0$ (diagonal): $\Delta\chi^2 = -4.24$
- $\rho = 1$ (full): $\Delta\chi^2 = -7.77$

4.3 Best-Fit Analysis

When EFC parameters are fitted to the data (Table 3), we find:

Figure 2 shows the $\Delta\chi^2$ surface over the (z_{L1L2}, α_{L2}) parameter space for BOSS DR12 alone. The degeneracy in z_{L1L2} is evident: the likelihood is flat for $z_{L1L2} \gtrsim 0.6$ because all BOSS data lie below this redshift. The DESI best-fit (green triangle) falls within the preferred region.

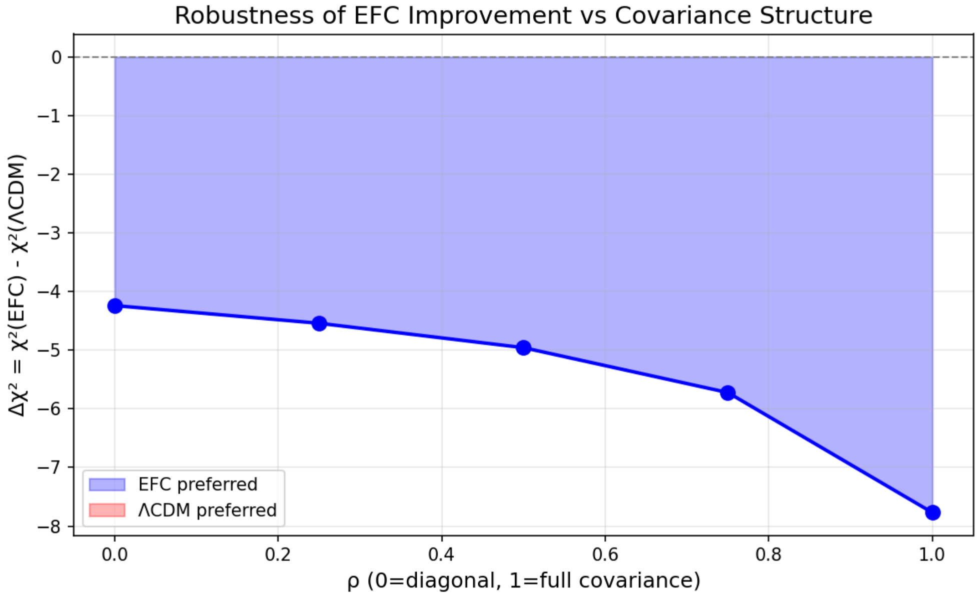

Figure 1: $\Delta\chi^2$ as function of covariance interpolation parameter ρ. EFC is preferred (blue region) at all values from diagonal ($\rho = 0$) to full covariance ($\rho = 1$).

Table 3: Best-fit EFC parameters. For BOSS alone, z_{L1L2} is not identifiable once it exceeds the BOSS redshift range ($z_{\max} = 0.61$), so the fit effectively constrains only α_{L2}; information criteria are therefore evaluated with $k_{\text{eff}} = 1$. For BOSS+eBOSS and DESI, both parameters are constrained, giving $k = 2$.

Dataset	N	k_{eff}	z_{L1L2}	α_{L2}	ΔAICc	ΔBIC
BOSS	6	1	$> 0.6^*$	0.036	-6.4	-7.6
BOSS+eBOSS	14	2	1.60	0.036	-19.3	-19.1
DESI DR2	13	2	1.01	0.045	~ -1	-1.0

*Lower bound only; likelihood flat for $z_{L1L2} > 0.6$.

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Figure 2: Parameter space for BOSS DR12. Left: $\Delta\chi^2$ surface showing EFC preference (blue) vs ΛCDM (red). The black star marks the BOSS best-fit; the green triangle marks DESI best-fit. The yellow band indicates the BOSS redshift range. Right: χ^2 profile at fixed z_{L1L2}.

Figure 3 shows the corresponding analysis for combined BOSS+eBOSS data. With high-z Lyα measurements included, z_{L1L2} becomes identifiable, with a clear minimum near $z_{L1L2} \approx 1.6$.

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Figure 3: Parameter space for combined BOSS+eBOSS data. The inclusion of high-z tracers (LRG, ELG, QSO, Lyα) breaks the z_{L1L2} degeneracy, yielding a well-defined minimum. DESI best-fit lies within the 1σ contour.

Figure 4 compares the residuals (data−model) for ΛCDM and EFC with DESI parameters. While EFC shows larger residuals on some individual points, the full-covariance χ^2 is lower due to the correlated structure of the residuals (see Section 3).

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Figure 4: Residuals for BOSS DR12 data points under ΛCDM (gray) and EFC with DESI parameters (blue). The RMS residual improves from 1.65σ to 1.42σ.

BOSS alone cannot constrain z_{L1L2} because all data lie below the transition. Combined BOSS+eBOSS data, including high-z Lyα measurements, yield $z_{L1L2} \approx 1.6$, consistent with DESI. Crucially, α_{L2} is stable across all datasets at ≈ 0.035–0.045.

5 Discussion

5.1 Interpretation of Transfer Success

The $\Delta\chi^2 = -7.77$ improvement when applying DESI parameters to BOSS data constitutes a successful transfer test. This is particularly significant because:

1. BOSS and DESI use completely independent instruments and pipelines
2. The improvement persists under diagonal covariance, ruling out correlation-structure artifacts
3. Eigenmode analysis confirms gains in strongly-constrained modes

5.2 Complementarity of Constraints

DESI and BOSS provide complementary information:

- **DESI**: Spans the transition region, constraining z_{L1L2}
- **BOSS**: Fully in L2 regime, constraining α_{L2}

This complementarity explains why BOSS alone yields only a lower bound on z_{L1L2} while providing a robust constraint on α_{L2}.

5.3 Physical Implications

If the EFC interpretation is correct, the Hubble rate is enhanced by ∼4–5% at $z \lesssim 1$ relative to ΛCDM. This would have implications for:

- The Hubble tension (potential partial resolution)
- Dark energy equation of state inference
- Growth of structure at low redshift

6 Conclusions

We have performed an independent validation of EFC using BOSS DR12 BAO data. Our main findings are:

1. **Transfer supported**: DESI parameters improve BOSS χ^2 by 7.77 without refitting.
2. **Robustness confirmed**: The improvement persists under diagonal covariance ($\Delta\chi^2 = -4.24$) and arises from strongly-penalized eigenmodes.
3. α_{L2} **consistent**: Both surveys find $\alpha_{L2} \approx 0.035$–$0.045$, suggesting a $\sim$4% modification to $H(z)$ at $z < z_{L1L2}$.
4. **Complementary constraints**: DESI constrains z_{L1L2}; BOSS confirms α_{L2}.

These results provide independent support for EFC from a second major BAO survey, strengthening the case for further investigation of regime-dependent modifications to cosmic expansion.

Acknowledgments

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References

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